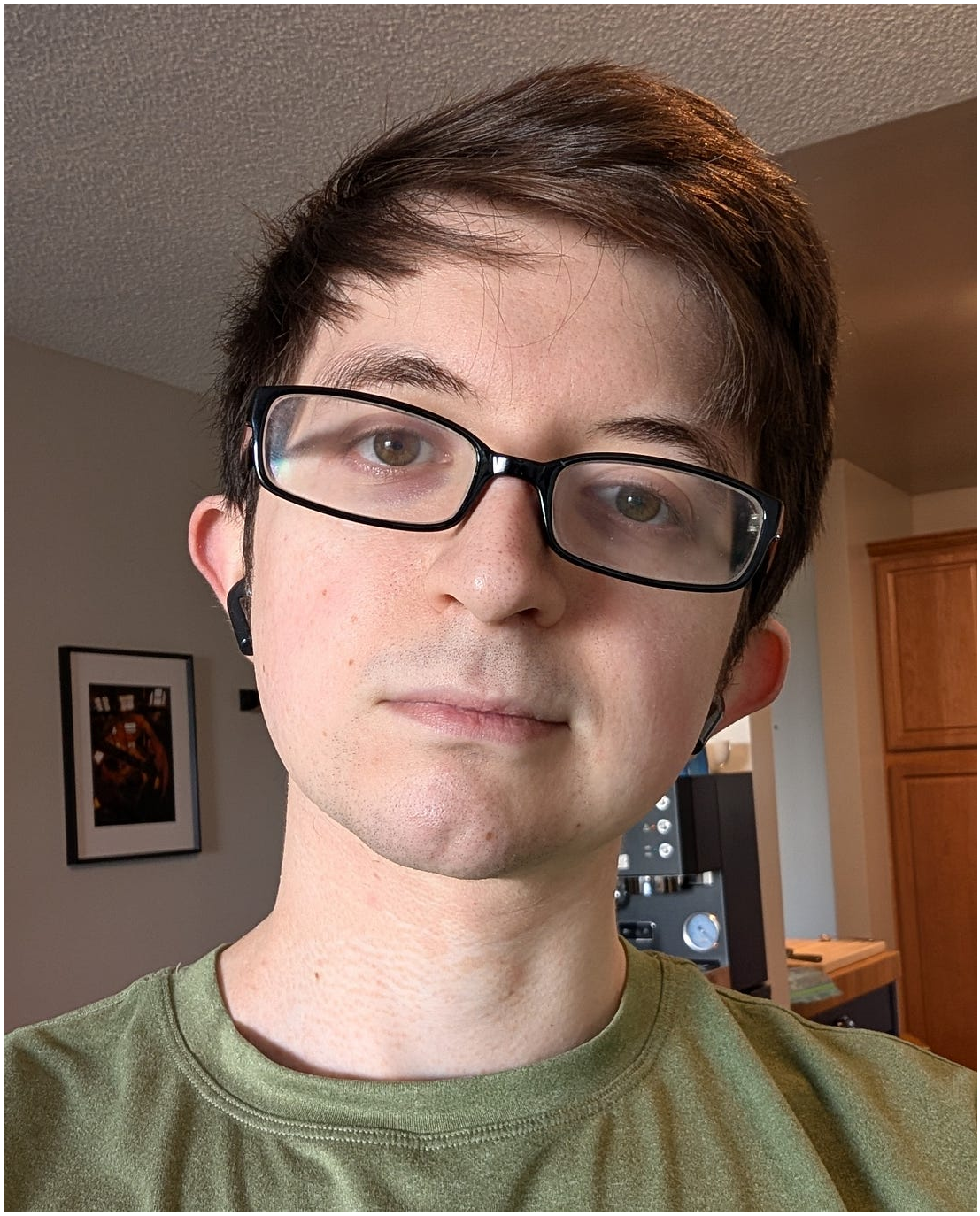
"Formalizing Buddhist Emptiness Without Losing Modus Tollens" - Chance Chapman (Independent Scholar)

Comparative Philosophy, 2026



MARCUS ARVAN

AUG 18, 2026



By Chance Chapman

Buddhist ‘emptiness’ is easy to mistake for the claim that nothing exists. The Indian Buddhist philosopher Nāgārjuna, operating sometime during the late 2nd and early 3rd centuries CE and traditionally held to be the founder of the “Madhyamaka” or “Middle Way” school of philosophical

Mahayana Buddhism, was preoccupied with defending emptiness dialectically as a coherent and intellectually persuasive doctrine. Madhyamaka forms the structural skeleton of Tibetan and Chan/Zen Buddhist thought; Nāgārjuna accordingly is a key figure of Mahayana Buddhism, second only to perhaps the Buddha himself. (Indeed, Nāgārjuna himself asserts that he expounds the teachings of the Buddha rather than forming an independent doctrine.)

Western scholarship has failed to come to a consensus on the character and classification of Nāgārjuna's argumentation and viewpoints, and interpretations have ranged from the Kantian (in which phenomenon is differentiated from an ultimately real noumenon) to the Wittgensteinian and even to assertions that Nāgārjuna is objectively an irrational thinker. Much of this difficulty revolves around the particular dialectical schema used by Nāgārjuna known as the 'tetralemma' or 'catuṣkoṭi.' While it is generally accepted that Nāgārjuna's mechanism of refutation is that of *modus tollens,* forming *reductio* attempts to render the opponent's position invalid from absurdity, the catuṣkoṭi does not seem to obey the rules of classical logic. It presents a matrix of four options:

> P (say, "Things fundamentally exist");
> Not-P ("Things fundamentally do not exist");
> Both P and Not-P ("Things both fundamentally and do not fundamentally exist");
> Neither P Nor Not-P ("Things neither fundamentally nor not-fundamentally exist").

(This sense of "fundamental" or intrinsic existence, known in Sanskrit as svabhāva, is the primary target of Nāgārjuna's critique.)

Moreover, Nāgārjuna never clearly appears to endorse any of the four. In fact, his use of *reductio* and *modus tollens* is invariably aimed at a rejection of the ultimate coherence – and thus, ultimate truth – of each position in order to affirm emptiness as the liberatory alternative. The general methodology of Nāgārjuna is to say that "if P were correct, Q

would also be correct; however, Q cannot possibly be correct, so P cannot be correct either," and so forth for each of the four arrangements.

A particular point of difficulty is the "no thesis" defense that Nāgārjuna outlines in his *Vigrahavyāvartanī* (The End of Disputes), written in response to critiques of his prior work, the *Mūlamadhyamakakārikā* (Root Verses of the Middle Way), in which Nāgārjuna claims:

> "29. If I had any thesis, that fault would apply to me. But I do not have any thesis, so there is indeed no fault for me (Westerhoff 2010, 30)."

At face value, this appears to be self-defeating; Nāgārjuna seems as if he's advancing a thesis of not having a thesis.

Just as many interpretations of Nāgārjuna's viewpoint itself exist, there exists many interpretations of the catuṣkoṭi. The most notable attempt within the realm of formal logic has been by influential logician Graham Priest, who has argued for a viewpoint informed by his own dialetheism in which paraconsistency, or true contradictions, exist; Priest therefore claims that Nāgārjuna's intent is to affirm the workability of true contradictions. Priest further posits that Nāgārjuna asserts a "fifth value" in which ultimate metaphysical distinctions are asserted as essentially ineffable.

However, dialetheist logic confronts a puzzle: Because dialetheism accepts "Both P and Not-P", unlike in classical logic, *modus tollens* therefore fails to be valid as an argumentative structure with determinative weight: if P implies Q, and Q is false, in a basic dialetheist interpretation P could be both true and false rather than simply false as determined within classical logic, robbing Nāgārjuna of logical weight. Priest has offered a solution to this that allows dialetheist logic to retain *modus tollens* but admits himself that it is extremely complex and "not, in general, effectively checkable."

In "The Modal Catuṣkoṭi: Formalizing Emptiness in Classical Modal Logic," I advance a construction and interpretation of the catuṣkoṭi that does not need exotic logical systems to preserve Nāgārjuna's use of *modus tollens*. Standard possible-worlds and modal operator frameworks are used to argue that what Nāgārjuna's really after is the *refutation of the necessity* of any assignment of a truth value to any proposition. This can be rendered into rather simple modal logic, with the single initial complication of the use of Belnap-Dunn semantics, developed in the 1970s to better allow computers to answer questions that involve incomplete or contradictory information. This is strikingly like the catuṣkoṭi already, in that it is a four-valued logic system that uses true (t), false (f), both (b) and neither (n).

The formula, T(P), is as follows:

> $T(P) =def \neg\Box(v(P) = t) \wedge \neg\Box(v(P) = f) \wedge \neg\Box(v(P) = b) \wedge \neg\Box(v(P) = n)$

Or something like: "No truth valuation of P is fixed as necessary."

Crucially, this applies to itself in a nested sort of fashion: the assertion "No truth valuation of P is fixed as necessary" *also does not have a necessarily assigned truth value*. Because T(P) is itself a formula, the same operation can be applied to it and then applied again. The syntax can operate in this manner to any finite depth. (For fans of Liar Paradoxes, this sidesteps such problems by being so nested; by analyzing the preceding formula rather than diagonally declaring itself false, no contradiction arises.)

This formula is additionally verified in Lean 4, a rigorous mechanical proof checker, under various systems of modal logic.

Articulating Nāgārjuna's thinking in this fashion allows us to sidestep a great deal of interpretive issues while remaining parsimonious and faithful to the method of implementation. For one thing, it's downright definitional that svabhāva, as a target of Nāgārjuna's refutation, involves

modal concepts like necessity, as otherwise svabhāva would not be speaking of something fundamental and intrinsic.

For another, it provides clear intelligibility to the thinker's various applications of *modus tollens*. The most cited example in the *Mūlamadhyamakakārikā* is that of an opponent who proposes that 'going' inherently exists in the path being 'gone' upon – i.e. that paths have an essential or inherent quality of 'going-ness'. With our formula as a lens, Nāgārjuna's argument acquires a straightforward logic: If a path *necessarily* possessed 'going-ness', someone walking on it to get somewhere – a 'goer', also *needing* to possess some kind of 'going-ness' – would *necessarily* entail two goers. But this is absurd, as then there would be two goers, and our sense faculties make clear that only one object is contingently in motion. As there are not two goers, possession of 'going-ness' cannot *necessarily* be the case.

(Note: Nāgārjuna's reasoning that a 'goer' would need to possess 'going-ness' in such an entailment is that if an opponent suggests a 'goer' need only interact or engage in 'going-ness' without possessing it as a property, that would mean the 'goer' would be interacting with the necessarily fundamental 'going-ness' in the path. However, that would make the svabhāva of 'going-ness' then go from a state of not being interacted with to a state of being interacted with – which would be a change to a necessarily fundamental existent, a contradiction; something that exhibits change to its properties no longer exhibits necessarily fundamental properties, and svabhāva would not make sense without these.)

As the formula iterates indefinitely upon itself (without necessarily supposing an infinitely coherent chain, only local coherence at each step), sustained application leads to an eventual recognition of the therapeutic characteristics of Buddhist philosophy, and of Nāgārjuna's 'no thesis' claim. He is clearly not asserting the opposite – that a path necessarily doesn't possess 'going-ness' - since that would trap him in the same bind. He is providing a methodological deconstruction of any

non-local, universal, and/or utterly non-contingent necessity claim, which is itself essentially what 'necessity' must tend toward being once we get to the realms of metaphysics, broad epistemics, et cetera.

Since the formula is Lean-verified as satisfiable, we consider that satisfiability correct in the sense that it operates coherently in classical logic and doesn't speak beyond it. Even tautologies display this distinction; for example, excluded middle is valid in classical logic but not First-Degree Entailment (FDE), where a proposition may be neither true nor false. Its classical validity can't certify classical logic as reality's uniquely necessary structure. (One may note this has a remarkably Gödelian air to itself.)

Because the formula operates under the rules of classical logic and binary negation, the initial (b) and (n) values are also not necessarily assigned; therefore, after the first application, it all boils down to basic and bivalent modal contingency: $T(\varphi) = \Diamond\varphi \wedge \Diamond\neg\varphi$. It's no wonder it checks out in Lean!

However, most uses of contingency are much more bounded rather than applying over any arbitrary P, and this is how Nāgārjuna's arguments begin to acquire non-foundationalist force. 'Any arbitrary proposition' is about as big of a topic as there is. To assert a claim of necessity in the first place – let alone a necessary metaphysics – one advances that claim as a proposition. A local necessity, like *modus tollens* in classical logic, is what we may call 'conventionally' or 'provisionally' necessary; it's a necessary consequence in classical logic, but not in dialetheist logic. This allows us to say things are necessary in this mundane, modest, *contingent* sense.

But a claim of fundamental, intrinsic, or 'ultimate' necessity has the same highest burden of proof within standard systems of logic as necessity itself does: All necessity needs to be refuted is the satisfiability of an admissible countermodel. Nāgārjuna is not playing by any special rules separate from that of the opponent. Therefore, if an opponent objects

that the ‘countermodel’ of Nāgārjuna’s argument is not admissible, Nāgārjuna need only ask what *fundamentally* bars the countermodel. As the satisfiability is otherwise valid within classical modal logic, the opponent invariably runs into either a bare posit of necessity that lacks strict justificatory or logical force, or an infinite regress that the opponent is unlikely to accept if they want to keep making sense of the word ‘fundamental’; i.e., foundational or ground-producing. Notably, for Nāgārjuna this regress becomes what I call virtuous rather than vicious; the indefinite iteration that accounts for itself at any local level of analysis is all that is needed for the job.

With the analysis concluded, it becomes clear that Nāgārjuna provides a rigorously coherent explanation of emptiness as a non-foundationalist philosophy of non-necessity, provisionally held as self-revisable, in which its consequences are argued to lead to the Buddhist path. In establishing that coherence, essentialist or necessitarian foundations are formally refuted within the bounds of classical modal logic.

Thanks for reading New Work in Philosophy!
Subscribe for free to receive new posts and
support my work.

Pledge your support